Luke Rodel

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EDUCATION

Bachelor of Science in Mechanical Engineering | The University of Alabama| Robotics Minor | 3.96 GPA | May 2027

EXPERIENCE

Mechanical Engineering Intern | Herrick Technology Laboratories | Germantown, MD | May 2026 - August 2026

- Repaired broken CAD models, drawings, and BOMs, standardizing the CAD database and preventing downstream manufacturing errors
- Established thermal testing standards, via physical testing and Autodesk CFD simulations, to validate and transition to thermally superior parts
- Extracted Altium board layouts to optimize mechanical packaging and performed SWAP-C calculations for RF assemblies
- Prepared hardware for environmental drop and immersion testing, and designed custom carrying cases with calculated foam inserts
- Developed Creo .hol files enabling one-click creation of threaded inserts and automatic drawing callouts, saving design time while reducing error

Industrial Engineering Co-Op | Mercedes-Benz U.S. International | Tuscaloosa, AL | May 2025 - August 2025

- Developed a PowerBI dashboard to perform Pareto analysis on quality issues, drastically reducing the time required to identify top scrap drivers
- Investigated data anomalies exposed by the dashboard, uncovering a fragmented process that concealed millions of dollars in untraced product, improper charge-backs, and excess re-ordering
- Formed a proposal for a unified scrap management system to close these financial gaps, correct inventory data, and automate the recovery process
- Championed AI integration within the department by identifying high-value automation opportunities, co-developing agents with the Data Team, and leading training sessions to modernize production workflows

Battery Series Planning Co-Op | Mercedes-Benz U.S. International | Tuscaloosa, AL | August 2024 - December 2024

- Designed and deployed an automated label printing and vision grading system, meeting regulations and reducing takt time by 10 seconds
- Developed an automated data comparison tool to validate error logs, eliminating thousands of manual data entry errors and saving countless hours of verification time
- Established a preventative maintenance protocol for High Voltage (HV) Caps, resulting in a measurable reduction in component defects
- Restructured production line workflows to maintain operational efficiency for running at half speed and half staff

Design Engineer Intern | Spudnik Equipment Company | Blackfoot, ID | May 2024 - August 2024

- Redesigned a 12-row Sugar Beet Harvester gear train to improve tensioning reliability and simplify assembly, reducing the number of parts by 10
- Executed over 100 design modifications (DFM) using NX, including slot/tab integration and adjusting dimensions to streamline fabrication
- Worked all manufacturing processes in a vertically integrated company, gaining valuable insight on design for tooling and assembly
- Analyzed regulatory safety standards to determine compliance requirements and integrated safety labeling directly into CAD models

Maintenance Engineering Co-Op | Mercedes-Benz U.S. International | Tuscaloosa, AL | January 2024 - May 2024

- Designed, fabricated, and mounted support brackets for warped front end mounts permanently fixing headlight/panel alignment quality issue
- Managed improvement projects for vibrational analysis, flow meters, and pit lighting, including drafting proposals and securing vendor bids
- Created FIFO rack for lineside roller replacement parts decreasing repair time, ensuring part availability, and reducing overstocking.
- Reverse Engineered/3D Modeled a multitude of broken parts enabling repairs and saving thousands in replacement parts versus purchasing

Electric Motor Repair Intern | Electric Motor Service Company | Idaho Falls, ID | June 2021 - August 2021

- Disassembled and cleaned broken motors, preparing them for repair by technicians
- Assembled new gearboxes and motors to fulfill orders, ensuring each one was built to specifications
- Performed precision balancing on serviced motor rotors, a critical step that significantly increases motor lifespan
- Prepared and painted repaired motors, ensuring a professional finish and protecting components from wear

PERSONAL PROJECTS

- **Senior Capstone:** Engineering a robotic arm Direct Ink Writing system and end effector capable of printing non-planar polymer lattices
- **Software & AI:** Deployed local AI models and agents, developed web apps, and completed intro courses in Python and PLC Programming
- **Systems & Networking:** Built personal computers and a home media server; ran Ethernet cabling and installed a security camera system
- **Welding & Shop Safety:** Completed a beginner's welding course at the College of Eastern Idaho and obtained an S/P2 safety certification
- **Residential Building:** Planned and built a wooden pergola-style carport as well as a large, brick, upright fireplace

TECHNICAL SKILLS

NX, SolidWorks, Creo, Autodesk CFD, PowerBI, MATLAB, Python, Docker, Welding, 3D Printing, and Machining